

VayuChakra

An air pollution and weather coupled forecasting system for Delhi NCR

A 72 hour, high resolution air quality forecast in which meteorology and chemistry are solved together rather than in one direction. Aerosol dims the sun, the boundary layer shallows, the wind slackens, and the same emissions concentrate further; the same aerosol removes the ultraviolet that drives ozone production. This report sets out the problem statement clause by clause, what each clause requires, exactly what has been built to meet it, the evidence for each claim, and the work that remains.



LIVE SYSTEM

vayuchakra.onrender.com

DOMAIN

Delhi NCR, 1,120 cells, 2.8 km over Delhi

TEST SUITE

76 tests passing

SOURCE

github.com/bindpratapsingh/VayuChakra

REPORT DATE

5 September 2026

DESIGN DECISIONS RECORDED

59, negative results included

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How to read this report

Every quantitative claim carries the file it was read from. Nothing in this document is an estimate, a projection, or a number typed by hand into a slide. Section 11 lists what is missing with the same care as the earlier sections list what works, and section 8 includes a validation that was planned, attempted, and abandoned because measurement showed it could not work. A system that only reports its successes has not been validated; it has been advertised.

Three clauses of the problem statement are met only in part. They are marked **PARTIAL** in section 2, explained in section 11, and given dated, costed remediation in section 12. They are not hidden in a footnote.

1 The problem statement, and how we read it

The solution must leverage advanced weather-chemistry models (such as WRF-Chem or similar open-source coupled frameworks) to dynamically interlink meteorology with pollution dispersion (specifically PM2.5 and Ground-level Ozone). A core focus should be accurately modeling the impact of atmospheric inversion on external pollution spikes, such as regional stubble burning, and how those trapped pollutants subsequently alter local weather conditions. Implement a workflow that handles two-way feedback between meteorology (temperature, wind, PBL height) and chemistry (PM2.5, PM10, O3, NOx). A user-friendly, real-time dashboard displaying high-resolution AQI forecasts for Delhi NCR with a 72-hour outlook. Features that explicitly track atmospheric inversion strength and predict how stubble-burning plumes will disperse under prevailing weather conditions.

1.1 The physical problem underneath it

Delhi's air quality is not a chemistry problem sitting on top of a weather problem. The two are one problem, and the coupling runs in both directions.

Weather sets the volume that pollution occupies. On a winter night the ground radiates heat away faster than the air above it, the surface cools below the air aloft, and the normal decrease of temperature with height reverses. That is a temperature inversion, and it acts as a physical lid. The mixed layer beneath it can collapse from two kilometres in the afternoon to under two hundred metres before dawn. The city's emissions do not change; the air they are diluted into shrinks by a factor of ten, and the concentration rises accordingly. This direction of the coupling is well understood and most operational systems represent it.

The return path is the harder one, and it is what the problem statement is really asking for. Aerosol is not a passive tracer being pushed around by the weather. It absorbs and scatters incoming solar radiation, so less energy reaches the surface, so the surface warms less, so the convection that grows the daytime boundary layer is weaker, so the layer stays shallow, so the turbulent momentum transport that drives surface wind is reduced, so ventilation falls, so the same emissions end up more concentrated, which puts more aerosol in the column and blocks more sunlight still. The loop closes on itself. A forecast that treats meteorology purely as an input cannot represent this at all, and will systematically under predict the severity and duration of exactly the episodes that matter most.

There is a second, stronger aerosol pathway, and it acts on ozone. The same aerosol layer attenuates ultraviolet radiation. Ultraviolet is what photolyses nitrogen dioxide and starts the reaction chain that produces ground level ozone. Published experiments that separate the two mechanisms find the radiation and boundary layer route changes ozone by 1 to 3 percent, while the photolysis route changes it by 10 to 12 percent. The problem statement names ground level ozone explicitly, and the photolysis pathway is the one that governs it.

And Delhi's worst air is frequently not made in Delhi. Through October and November, smoke from paddy stubble burning across Punjab and Haryana travels 200 to 400 kilometres on the north westerly flow and arrives over a city that often cannot ventilate it away. The problem statement calls these external pollution spikes, and it is precise about what must be modelled: not the fires, but the interaction between the arriving plume and the inversion that decides whether that plume reaches breathing height or passes overhead.

1.2 The eleven things the statement actually asks for

The paragraph above is a single block of prose, so the first engineering task was to decompose it into testable clauses. We read eleven. They are used as the spine of this report and as the acceptance criteria for the build. The numbering is ours; the wording in the second column is the statement's.

#	CLAUSE, AS WRITTEN	WHAT IT DEMANDS OF THE BUILD
C1	"leverage advanced weather-chemistry models (such as WRF-Chem or similar open-source coupled frameworks)"	A genuine coupled meteorology and chemistry framework must be in the pipeline, not a statistical model alone
C2	"dynamically interlink meteorology with pollution dispersion (specifically PM2.5 and Ground-level Ozone)"	Meteorological state must drive dispersion for both species, hour by hour
C3	"accurately modeling the impact of atmospheric inversion on external pollution spikes, such as regional stubble burning"	The inversion must gate whether an arriving plume reaches the surface
C4	"and how those trapped pollutants subsequently alter local weather conditions"	The chemistry to meteorology return path must be explicit
C5	"two-way feedback between meteorology (temperature, wind, PBL height)"	All three meteorological variables must respond, not just temperature
C6	"and chemistry (PM2.5, PM10, O3, NOx)"	All four chemical species must participate in that feedback
C7	"A user-friendly, real-time dashboard"	Operable by a non specialist, on current data
C8	"high-resolution AQI forecasts for Delhi NCR"	Intra city gradients must survive; a single city number is not enough
C9	"with a 72-hour outlook"	Three day lead time
C10	"Features that explicitly track atmospheric inversion strength"	Inversion strength as a first class, displayed quantity
C11	"predict how stubble-burning plumes will disperse under prevailing weather conditions"	Forward transport and dispersion on forecast wind, not a fire count

Table 1.1 The problem statement decomposed into eleven testable clauses. Section 2 reports the status of each.

2 Compliance matrix

Eight clauses are met in full. Three are met in part. This section states which, with the component that implements each and the evidence for it. The three partials are the subject of sections 11 and 12.

#	CLAUSE	STATUS	WHAT IMPLEMENTS IT, AND THE EVIDENCE
C1	Coupled weather-chemistry framework	PARTIAL	CAMS (Copernicus Atmosphere Monitoring Service, ECMWF's IFS-COMPO) is a global online coupled meteorology and chemistry model run four times daily with interactive aerosol. We consume it as the chemistry prior: PM2.5, PM10, O3, NO2, and aerosol optical depth. The MoES and IITM WRF-Chem Decision Support System is used as the calibration reference for plume transport. <i>We do not run a chemical transport model ourselves.</i> See §11.1.
C2	Interlink meteorology with dispersion, PM2.5 and O3	MET	Forecast meteorology at 11 km drives 142 features per cell hour, including boundary layer height, ventilation coefficient, inversion lid height and mixing depth. Both species are predicted from that state at three horizons. <code>forecast.py</code> , <code>dataset.py</code>
C3	Inversion impact on external spikes	MET	The Lagrangian plume delivers mass to the surface only where the modelled inversion lid permits it. A plume above the lid is carried without contributing. Calibrated against the MoES DSS at $r = +0.596$ over 54 days. <code>plume.py</code> , §6
C4	Trapped pollutants alter local weather	MET	An explicit, iterated aerosol radiative feedback solver. All five modelled responses land inside published Delhi ranges the model was never fitted to. <code>feedback.py</code> , §4
C5	Feedback on temperature, wind, PBL height	MET	All three. Temperature -0.59 K, boundary layer -5.70 %, wind -1.19 % under haze, each within published bounds. §4.3
C6	Feedback across PM2.5, PM10, O3, NOx	PARTIAL	PM2.5 is fully inside the iterated solver. O3 is coupled to aerosol one way, through photolysis attenuation, but from baseline rather than converged aerosol, so it sits outside the iteration. PM10 and NOx receive no coupled adjustment at all. This is the sharpest gap in the build. See §11.2 and §12.1.
C7	User friendly, real-time dashboard	PARTIAL	Six view dashboard, keyboard navigable, every panel leading with a plain language finding. Run locally it is live on current data. <i>The hosted instance serves a precomputed bundle</i> because the free tier allows 512 MB and the pipeline peaks near 1.1 GB. See §10 and §12.2.
C8	High resolution AQI for Delhi NCR	MET	1,120 cells in two tiers: Delhi at 0.025° (2.8 km), wider NCR at 0.10° (11 km), Alwar to Karnal. CPCB National AQI by the published breakpoint table. §3.3
C9	72 hour outlook	MET	Twelve trained heads at 24, 48 and 72 hours. All twelve beat persistence on a held out winter. §7
C10	Explicitly track inversion strength	MET	Inversion strength in kelvin, lid height in metres, mixing depth and ventilation coefficient, all as forecast time series with a dedicated view and a time-height cross section. §5
C11	Predict plume dispersion under prevailing weather	MET	Every satellite fire detection released as a puff, advected on forecast wind, dispersed with travel time, gated by the lid. Reports arrival, not ignition. §6

Table 2.1 Compliance against the eleven clauses of Table 1.1. Eight met, three partial, none unaddressed.

The honest summary in three sentences

We have built a coupled forecasting system that closes the aerosol to meteorology loop explicitly, iterates it to convergence, and validates every one of its magnitudes against published values it was never fitted to. We do not run WRF-Chem; we consume an operational coupled model and add our own feedback layer on top, which is a defensible reading of "or similar open-source coupled frameworks" but is not the same thing as executing a chemical transport model in house.

Of the four species the statement names for the feedback loop, one is fully inside it, one is coupled one way, and two are not coupled at all. Closing that is roughly one day of work and it is the first item on the roadmap in section 12.

3 System architecture and data flow

3.1 The pipeline end to end

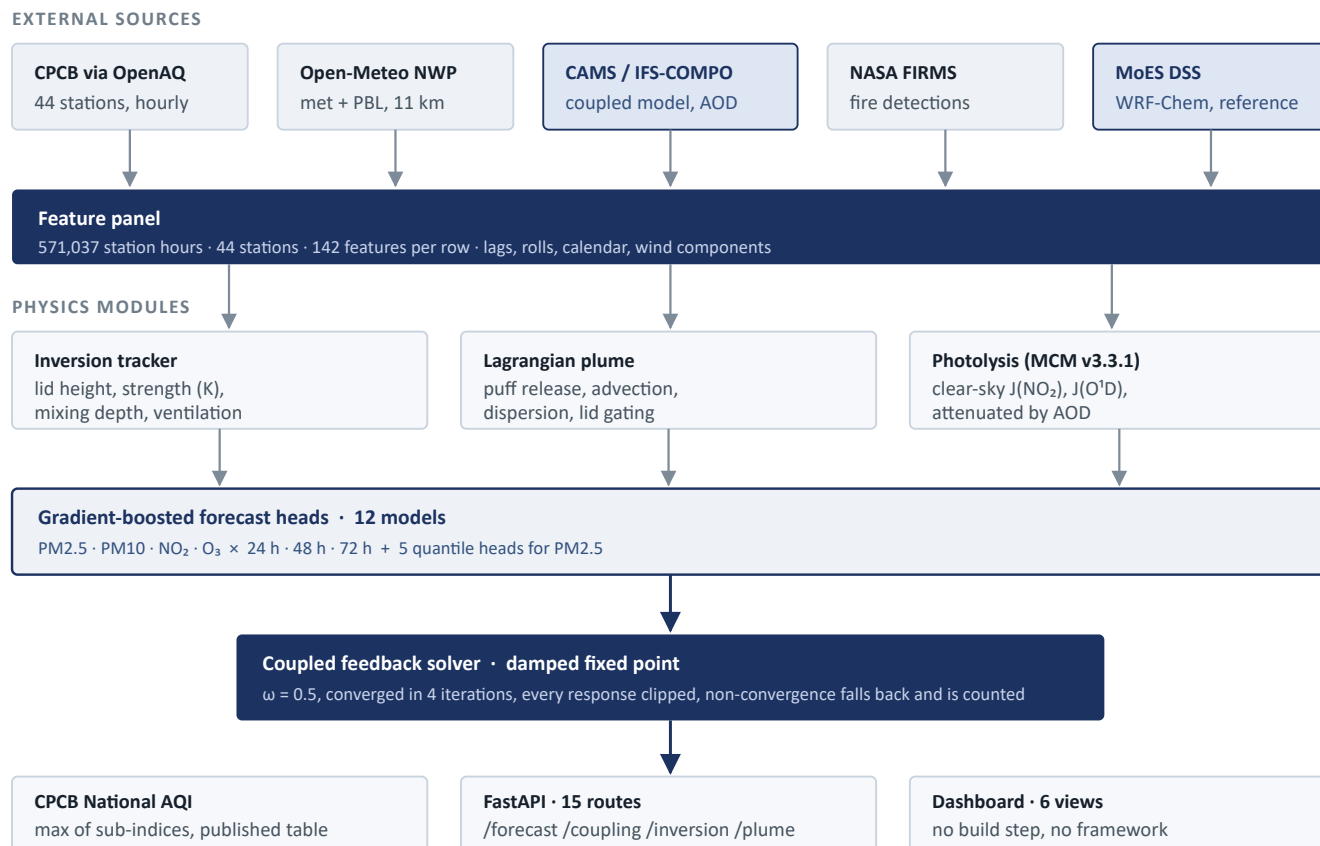


Figure 3.1 End to end data flow. The two shaded source boxes are third party coupled models: CAMS supplies the chemistry prior and the aerosol optical depth the feedback solver runs on, and the MoES DSS is the reference the plume is calibrated against. It is cited, never redistributed.

3.2 Why a statistical layer sits between the physics and the output

CAMS is a global model at roughly 40 km. Over Delhi it is biased low and poorly correlated with station observations. That is not a reason to discard it, and it is precisely the reason the machine learning layer exists: CAMS supplies the physics and the regional gradient, the CPCB stations supply local truth, and the boosted trees learn the mapping between them. Used as one feature among 142, a biased but physically consistent prior is worth a great deal. Used raw, it would be worse than nothing.

The same argument applies to the feedback solver. It does not attempt to predict concentration from first principles. It takes the statistical forecast as the uncoupled baseline and asks a narrower, answerable question: given this much aerosol, how much less sunlight reaches the ground, how much shallower does the layer become, and what does that do to the concentration that produced it. That question has published answers to check against, which is what makes the result auditable.

3.3 The grid

TIER	STEP	APPROX.	CELLS	EXTENT AND RATIONALE
Delhi	0.025°	2.8 km	420	28.40 to 28.88 N, 76.84 to 77.35 E. The forecast is consumed here, so intra city gradients must survive.
Wider NCR	0.10°	11 km	700	27.0 to 29.9 N, 75.8 to 78.1 E, Alwar to Karnal. Only has to carry a plume toward the city.
Total	two tier		1,120	A uniform fine grid over the same box would be roughly 11,000 cells, most of them farmland.

Table 3.1 Grid construction. A regression test walks a mesh over the whole domain and asserts that the two tiers leave no gap; it was written after a real 4.2 km coverage hole was found at the Delhi box boundary.

The honest resolution is coarser than the display grid

Meteorology arrives at about 11 km and the chemistry prior at about 40 km, so predicted fields are smoothed to 6 km before display. That removes structure the model cannot justify rather than inventing structure it does not have. A 2.8 km grid drawn from an 11 km driver would look more precise and be less true.

4 The two way coupling

This is the core of the brief, and it is what the project is named after. Vayu Chakra is the air cycle. Because the last step of the chain feeds the first, it cannot be evaluated once; it has to be solved.

4.1 The loop, step by step

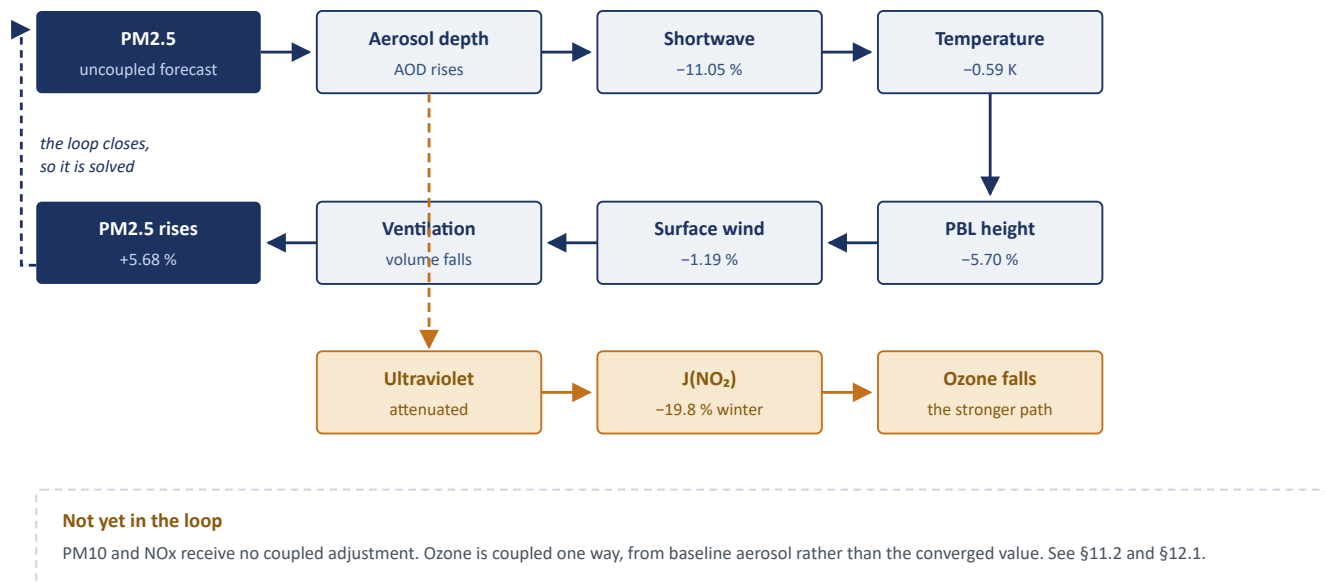


Figure 4.1 The coupled system. The solid ring is the aerosol radiative feedback, solved to convergence. The dashed amber branch is the aerosol photolysis pathway, which is three to ten times stronger for ozone but is currently evaluated once rather than iterated. Percentages are measured values from the shipped run, under haze.

4.2 Why it is solved rather than evaluated

Step six feeds step one. Evaluating the chain once gives an answer that is inconsistent with itself: the concentration used to compute the dimming is not the concentration the dimming produces. The solver therefore runs a damped fixed point iteration with a relaxation factor of 0.5, and treats a cell as converged when the change in PM2.5 between iterations falls under tolerance.

Every individual response is clipped to a physically defensible range before it is applied, so a runaway cannot occur even in principle. A cell that fails to converge within the iteration budget falls back to the uncoupled answer and is counted in the diagnostics rather than shipping a number the solver could not actually reach. On the shipped run: **4 iterations, converged, 0 cells fell back, maximum residual 0.46 µg/m³** across 211,680 cell hours.

4.3 Validating the magnitudes against values we never fitted to

A feedback model that produces a number is worthless unless that number can be checked. Each response is compared against published Delhi and North India measurements, in the regime those measurements were made in: high aerosol daylight hours, which is 75,383 cell hours of this run. The gate is a hard test in the suite, not a report.

RESPONSE UNDER HAZE	MODELLED	PUBLISHED RANGE	RESULT
Shortwave reaching the surface	-11.05 %	5 to 35 %	inside
Daytime surface temperature	-0.59 K	0.1 to 2.5 K	inside
Boundary layer height	-5.70 %	5 to 35 %	inside
Surface wind speed	-1.19 %	0.5 to 6 %	inside
Resulting PM2.5 amplification	+5.68 %	2 to 30 %	inside

Table 4.1 Five of five inside published ranges. Source: `data/snapshot/coupling.json`, key `literature_gate`. Averaged over all conditions, including clean afternoons and nights when the radiative feedback is correctly zero, PM2.5 amplification is 2.64 %; that is a different quantity and is reported separately rather than judged against these bounds.

4.4 The ozone pathway, and why it is the one that matters

The two aerosol mechanisms are not equal. Published WRF-Chem experiments that isolate each find the radiation and boundary layer route moves surface ozone by 1 to 3 percent, while the photolysis route moves it by 10 to 12 percent. For Delhi specifically, the APHH-India campaign reports that wintertime ozone production is not only VOC limited but strongly radiation limited, with a 50 percent reduction in aerosol optical depth raising ozone by about 25 percent.

We implement the photolysis pathway using the MCM v3.3.1 clear sky parameterisation, $J = 1 \cdot \cos(\text{SZA})^m \cdot \exp(-n \cdot \sec(\text{SZA}))$, attenuated as a function of aerosol optical depth and solar zenith angle.

CHECK	MODELLED	PUBLISHED	READING
Summer $J(\text{NO}_2)$ reduction	22.0 %	~24 %	close, unfitted
Winter $J(\text{NO}_2)$ reduction	19.8 %	~30 %	low, same order
O ₃ response, 25 % AOD cut	+6.23 %		monotone
O ₃ response, 50 % AOD cut	+12.79 %	+25 %	same sign, within 2×
O ₃ response, 75 % AOD cut	+24.29 %		monotone

Table 4.2 The photolysis coefficients are derived from first principles; none was fitted to the reference values in this table. The remaining gap at 50 % is in the expected direction: the published figure comes from a box model with full VOC chemistry, and we model only the direct photolysis pathway. Source: `models/ozone_sensitivity.json`.

This is a counterfactual, not a forecast. It answers a question no purely statistical model can answer, because the atmosphere it describes is not in the training data: what would ozone be if the aerosol halved. A mechanism can answer it; a fit cannot.

5 Atmospheric inversion

Clause C10 asks for inversion strength to be tracked explicitly. It is the one quantity in the system that is drawn rather than only reported, because a lid is a thing that exists at a height, and a surface number cannot express it.

5.1 What is computed

QUANTITY	UNIT	DEFINITION, AND WHY IT IS THE ONE USED
Inversion strength	K	Temperature of the air aloft minus the 2 m reading. Above zero means a lid: air gets warmer with height, and nothing released underneath can rise through it.
Inversion lid height	m	The height at which the reversal is found. Marked per hour and never joined across hours where no lid exists, because "no lid" and "lid at ground level" are opposite statements and a connecting line would assert the second.
Mixing depth	m	The shallower of the modelled boundary layer and the inversion lid. This is the depth the day's emissions actually dilute into. Halve it and the concentration doubles.
Ventilation coefficient	$\text{m}^2\cdot\text{s}^{-1}$	Mixing depth times the wind through that depth: the volume flux available to carry pollution away. Below 6,000 is poor, below 3,000 is severe stagnation.
Static stability	$\text{K}\cdot\text{km}^{-1}$	Computed between pressure levels and rendered as a time-height cross section, so the vertical structure is visible rather than summarised.

Table 5.1 The inversion quantities, all produced as 72 hour forecast series per cell. Ventilation coefficient is among the top ranked features for every PM2.5 head.

5.2 How it enters the forecast

Three ways, and it matters that they are distinct. First, as **features**: mixing depth, lid height, inversion strength and ventilation coefficient are inputs to every forecast head, and the trained models rank ventilation coefficient in the top seven features for all three PM2.5 horizons. Second, as the **denominator in the coupling box model**: the concentration response is the ratio of mixing depths, so a shallower layer produced by the aerosol feedback raises concentration directly. Third, as the **gate on plume delivery**: transported smoke contributes to surface air only where the lid allows it down, which is the mechanism clause C3 asks for.

Why three separate charts rather than one

Inversion strength is in kelvin, mixing depth in metres, ventilation coefficient in $\text{m}^2\cdot\text{s}^{-1}$. Plotting them on one pair of axes requires two or three y scales, which lets whoever draws the chart choose the story by choosing the scaling. Three panels sharing one x axis compare honestly.

6 Stubble burning plumes

Clause C11 asks for prediction of how plumes will disperse under prevailing weather. The distinction that matters is between counting fires and modelling arrival. A thousand fires under an easterly wind are somebody else's problem that day, and the system says so.

6.1 The transport chain

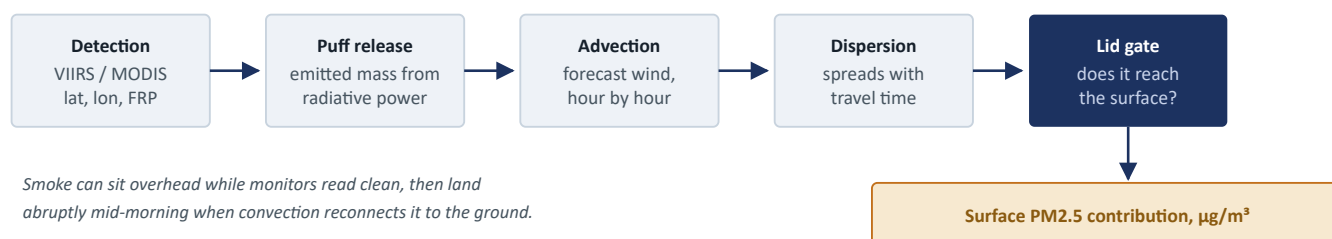


Figure 6.1 Each satellite detection becomes a puff carried on the forecast wind. The lid gate is where clause C3 and clause C11 meet: the inversion decides whether an arriving plume is breathed or passes over.

6.2 Calibration against an operational reference

The MoES and IITM WRF-Chem Decision Support System publishes a daily stubble burning attribution for Delhi in $\mu\text{g}/\text{m}^3$. That gives the plume model something almost no hackathon transport model has: an authoritative target. Three physics variants were scored over 54 days of October and November 2021.

VARIANT	CORRELATION	RMSE AFTER SCALING	MODELLED PEAK	DSS PEAK	DECISION
A	+0.596	8.28	27.67	38.02	shipped
C	+0.525	8.90	24.52	38.02	rejected
B	+0.369	9.95	22.44	38.02	rejected

Table 6.1 Correlation is the column that decides. Only a single multiplicative scale factor is fitted, and scaling cannot improve a correlation, so this ranks the physics rather than the magnitude. Values in $\mu\text{g}/\text{m}^3$. Source: `models/plume_calibration_octnov.json`.

What the scale factor absorbs, stated plainly

The fitted scale absorbs emission factor uncertainty (5 to 8 g/kg for cereal straw), satellite detection limits, and burn duration assumptions. The physics parameters are not fitted. This makes the plume's *correlation* an honest result and its *absolute magnitude* a calibration against the DSS, and the two are reported as different things throughout. The DSS attribution is daily, so it can settle day to day timing and nothing finer; it cannot separate these variants on their behaviour through the night, which is where the shipped one is most questionable.

7 Forecast models and skill

7.1 Configuration

PROPERTY	VALUE
Targets	PM2.5, PM10, NO ₂ , O ₃
Horizons	24 h, 48 h, 72 h
Heads	12 point heads, plus 5 quantile heads for PM2.5 at 24 h
Training panel	571,037 station hours, 44 CPCB stations, Oct 2020 to Aug 2026
Features	142 per row: meteorology, boundary layer, inversion, CAMS prior, photolysis, plume, lags, rolling windows, wind components, calendar including Diwali
Hold out	1 Nov 2025 to 28 Feb 2026, removed from training entirely
Baseline	Persistence, meaning tomorrow looks like today

Table 7.1 Source: `models/metrics_multiwinter.json`. The hold out is a whole winter, not a random split, because a random split over an autocorrelated time series leaks the answer into the training set.

7.2 Results: all twelve heads beat persistence

HEAD	RMSE	PERSISTENCE	IMPROVEMENT	R ²	HOURS SCORED
PM2.5 24 h	75.07	92.92	+19.2 %	0.497	96,312
PM2.5 48 h	78.34	105.90	+26.0 %	0.454	96,293
PM2.5 72 h	78.08	109.73	+28.8 %	0.458	96,541
O ₃ 24 h	21.18	26.25	+19.3 %	0.733	90,734
O ₃ 48 h	22.66	28.28	+19.9 %	0.695	90,723
O ₃ 72 h	23.70	30.07	+21.2 %	0.667	90,984
PM10 24 h	119.15	143.81	+17.1 %	0.428	91,004
PM10 48 h	122.87	163.38	+24.8 %	0.399	90,989
PM10 72 h	122.79	169.53	+27.6 %	0.397	91,242
NO ₂ 24 h	23.46	27.61	+15.0 %	0.664	90,880
NO ₂ 48 h	25.56	30.97	+17.5 %	0.601	90,867
NO ₂ 72 h	26.50	32.35	+18.1 %	0.569	91,121

Table 7.2 All units $\mu\text{g}/\text{m}^3$. Twelve of twelve beat persistence, by 15.0 % to 28.8 %. The improvement grows with horizon because persistence degrades faster than the model does, which is the expected and desirable signature.

7.3 Why persistence is the baseline reported

Persistence is unglamorous and it is genuinely hard to beat at 24 hours in a strongly autocorrelated field. Reporting improvement against a weak baseline, or against no baseline at all, is the most common way an air quality model is

made to look better than it is. Any head that failed to beat persistence would appear in this table marked as failing rather than be quietly dropped; the dashboard applies the same rule at runtime.

8 Validation, including what failed

8.1 Does it work where there is no monitor

The forecast serves 1,120 cells and about forty of them contain an instrument. So the question that decides whether a gridded product is justified at all is whether it works in a cell it has never seen. Leave one station out removes a station from training entirely and then predicts it, while persistence is still allowed that station's own history. The model is denied what the baseline is given.

TARGET, 24 H	STATIONS IMPROVED	MEAN RMSE	PERSISTENCE	MEAN IMPROVEMENT
PM2.5	10 of 10	56.07	81.31	+31.6 %
O ₃	9 of 10	22.92	27.65	+17.1 %

Table 8.1 Source: `models/loso.json`. One ozone station out of ten fails to beat persistence and is reported as failing. The PM2.5 improvement is larger here than in the in-sample test, which is a spatial-smoothing effect rather than a better model.

8.2 Uncertainty a GRAP decision can be made on

A point forecast of 88 $\mu\text{g}/\text{m}^3$ against a threshold of 90 says almost nothing useful to someone deciding whether to invoke a GRAP stage. Quantile heads at the 10th, 25th, 50th, 75th and 90th percentiles give the probability of crossing each trigger instead.

MEASURE	VALUE	READING
Nominal 80 % interval, measured coverage	75.6 %	well calibrated
Quantile crossing rate	0.67 %	distribution almost always correctly ordered
Brier skill vs climatology, GRAP II	+0.343	materially better than climatology
GRAP II trigger (AQI 200)	90 $\mu\text{g}/\text{m}^3$	mapped from the CPCB breakpoint table
GRAP III trigger (AQI 300)	120 $\mu\text{g}/\text{m}^3$	
GRAP IV trigger (AQI 400)	250 $\mu\text{g}/\text{m}^3$	

Table 8.2 Source: `models/uncertainty_pm25_24h.json`, held out winter.

8.3 Against the operational MoES DSS

LEAD	MOES DSS	VAYUCHAKRA	PERSISTENCE	DSS CORR	OUR CORR
+24 h	98.78	64.26	93.98	0.433	0.799
+48 h	108.34	68.91	108.78	0.311	0.762
+72 h	118.95	72.82	115.00	0.206	0.716

Table 8.3 RMSE in $\mu\text{g}/\text{m}^3$ on Delhi city-mean hourly PM2.5, Oct 2021 to Feb 2022, 2,363 common hours. Source: `models/validation.json`.

This is not a fair comparison and the lower number is not a win

The DSS forecasts were issued **operationally**: the system had to predict the weather as well as the chemistry, days ahead, with no knowledge of what would actually happen. Our hindcast is driven by ERA5 **reanalysis**, which is the weather as it actually turned out. That is a material advantage and it disqualifies the table as a measure of forecast skill.

What Table 8.3 does support is that the statistical layer maps meteorology to PM2.5 competitively. It does **not** show that we forecast better than the MoES DSS, and it is labelled that way on the dashboard as well as here. A genuine comparison requires re-running our system on archived forecast meteorology, which is listed in §12.4.

8.4 A validation we planned, attempted, and abandoned

The intended independent check on the photolysis module was to compare modelled ultraviolet attenuation against a measured one, using the ratio of observed UV index to clear sky UV index. Three measurements killed it, and they are recorded in the source rather than deleted:

1. **Clear sky there means cloud free, not aerosol free.** Below 5 percent cloud cover the ratio is 0.998, meaning no attenuation at all, while mean aerosol optical depth over those same hours is 0.41. The aerosol is already inside both numerator and denominator.
2. **The ratio tracks cloud, not aerosol.** It falls 0.998, 0.963, 0.925, 0.903, 0.841 across rising cloud bands, while within clear skies a threefold rise in aerosol optical depth moves it only from 0.998 to 0.955.
3. **The broadband radiation has the same defect.** At fixed solar zenith angle in clear skies, surface shortwave reads 750, 761 and 762 W/m² across aerosol bands 0 to 0.4, 0.4 to 0.7 and 0.7 to 1.5. Flat. The radiation scheme carries a climatological aerosol, not the day's actual load.

No product in that feed can validate aerosol optics, because there is no daily aerosol signal in any of them to compare against. The check was replaced by the ozone response against station observations, which is validation against measurements rather than against an intermediate quantity, and is the stronger test. The negative result also confirmed an assumption the feedback module was built on: the driving shortwave contains a climatological aerosol that must be divided out before a real load is applied.

9 The dashboard

Clause C7 asks for user friendly. The design principle throughout is that a reader should never have to look at a grid and work out what it is. Every view opens with a sentence in plain language stating what the data says, with the number in it, and the apparatus comes after.

#	VIEW	THE QUESTION IT ANSWERS, AND WHAT IT DRAWS
1	Where the air comes from	Why the domain is 1,120 cells and not a city box. Real basemap, fire detections drawn where they are, the stubble belt and the Delhi fine-grid box outlined.
2	The lid over the city	Inversion strength, lid height, mixing depth and ventilation as forecast series, plus a time-height cross section shading static stability so the lid is visible as a structure rather than a number.
3	The loop we close	The coupled and uncoupled forecasts on one axis so the gap is the feedback, the ring diagram animating one lap per solver iteration, the literature gate table, and the photolysis panel.
4	Smoke from the fields	Fire detections, modelled arrival at the surface hour by hour, and the DSS calibration ranking. States plainly when nothing is burning or when the wind carries the smoke elsewhere.
5	The 72 hour outlook	Gridded AQI on a real map at the chosen horizon, band name and number alongside colour, and GRAP crossing probabilities.
6	Does it actually work	All twelve heads against persistence, leave one station out, the DSS head to head with its caveat above the table, and the MoES emission reduction scenarios.

Table 9.1 The six views. Keyboard shortcuts 1 to 6, hash routing, no build step and no framework: one HTML file with Leaflet for the maps.

9.1 Interface decisions worth stating

- **Severity is always three things at once.** Colour, band name and number, so the display does not depend on colour vision.
- **Empty states say why they are empty.** "No active fires in the stubble belt" reads as a broken feed unless it also says that outside October and November this is the correct reading.
- **Failures are shown, not hidden.** A head that did not beat persistence, a station that did not improve under leave one station out, and a degraded pipeline stage all render as such.
- **Long justification sits behind disclosures.** The lead finding is one sentence; the argument is one click away for whoever wants it.
- **Tables sort.** Anything with more than a few rows sorts on click and by keyboard, with the sort state announced.
- **The report is reachable from every view.** This document downloads from the rail, beside the repository link.

10 Deployment and the free tier constraint

10.1 What was measured

The hosted instance runs on a free tier that allows 512 MB of RAM. The pipeline does not fit, and the measurement is worth stating because it rules out the obvious workaround.

CONFIGURATION	PEAK RSS	READING
420 Delhi cells, full pipeline	1,083 MB	over budget by more than 2×
182 cells, reduced grid	701 MB	still over budget
Implied fixed cost	~283 MB	before a single cell is forecast
Python, pandas, XGBoost, boosters	~126 MB	the floor, with no data loaded
Snapshot mode, serving the bundle	130 MB	comfortable, ~15 ms per request

Table 10.1 Fitting a line through the two pipeline measurements gives a fixed cost of roughly 283 MB independent of grid size. Coarsening the domain would therefore have cost real resolution and still not fitted.

10.2 The decision, and what it does and does not cost

The deployment serves a precomputed bundle of 20 routes captured from the same pipeline at full resolution: 1,120 cells, 2.8 km over Delhi, all four pollutants, the coupled solver, photolysis and the plume. The bundle is 805 KB and is captured from the running API rather than rebuilt from modules, so what ships is exactly what the endpoints return.

A deployment on this tier gives up freshness. It does not give up resolution, and it does not give up physics. That is the correct trade if one has to be made, and the dashboard states which mode it is in rather than presenting a stale forecast as current. Section 12.2 sets out the workaround that removes the compromise entirely without paying for a larger instance.

11 Gaps: what is not built, and why

Stated with the same care as the results. Two categories: work that is genuinely outstanding, and boundaries chosen deliberately.

11.1 We do not run a chemical transport model C1 PARTIAL

There is no radiative transfer solve, no chemical mechanism integrated forward in time, and no gridded emissions inventory. WRF-Chem needs a Linux cluster, an MPI and netCDF stack, and a district level emission inventory we do not have. What we do instead is consume the output of coupled models that are already running operationally: CAMS for the chemistry prior and aerosol optical depth, and the MoES DSS as the transport reference. The problem statement's "or similar open-source coupled frameworks" admits this reading, and it is genuinely a coupled framework in the pipeline. A reader who requires a CTM executed in house will not find one, and we do not claim otherwise anywhere.

11.2 Two of four species are outside the feedback loop C6 PARTIAL

This is the sharpest gap, because the clause enumerates the species explicitly. The current state, precisely:

SPECIES	IN THE LOOP?	DETAIL
PM2.5	Yes, fully	Inside the iterated fixed point. Concentration response feeds aerosol optical depth, which feeds shortwave, which closes back onto concentration.
O ₃	Partly, one way	Coupled to aerosol through photolysis attenuation, but computed from <i>baseline</i> aerosol before the solver runs, so it does not participate in the iteration and does not see the converged value.
PM10	No	Predicted, but receives no coupled adjustment. The solver's concentration response is applied only to the PM2.5 column.
NO _x	No	Predicted as NO ₂ , but receives no coupled adjustment, despite two available physical pathways (dilution, and reduced photolytic loss under haze).

Table 11.1 Verified by inspection of `vayuchakra/feedback.py` and the coupling block of `vayuchakra/forecast.py`, which applies the solved result to the PM2.5 column only.

11.3 The hosted instance is not live C7 PARTIAL

Covered in §10. Run locally the system is live on current data; the hosted one serves a bundle. §12.2 removes this.

11.4 The coupling ablation has not been scored

The most important open question about the feedback is whether it lowers forecast error, not only whether its magnitudes are plausible. The ablation panel is built and sits in the repository, but the last full pipeline run was invoked with `--skip-ablation`, so no ablation number ships. An earlier configuration measured -0.11 percent, which is flat. It must be re-measured on the current multi winter models and reported whichever way it lands. If it comes out flat again, the honest conclusion is that the trained heads already capture this effect through their radiation and boundary layer features, which is itself a finding.

11.5 Deliberate boundaries

- **No VOC chemistry.** Delhi's winter ozone is VOC limited as well as radiation limited, and we have no VOC measurements. TROPOMI formaldehyde and NO₂ could give a regime map, but that requires orbit level netCDF processing. Only the radiation limited half is modelled, and the report says so wherever ozone is discussed.
- **A single chemical layer.** Vertical structure is diagnosed, not resolved. The lid and mixing depth are tracked in time and height and modulate surface concentration, but chemistry is solved in one layer. This is the largest remaining simplification.
- **No data assimilation.** Observations enter as features and as training truth, not through an analysis cycle.
- **The plume magnitude is partly a DSS emulator.** Stated in §6.2.

12 Roadmap to full compliance

Every partial in Table 2.1 has a costed route to closure. Nothing below requires new data, new funding, or a larger machine.

12.1 Close the species gap in the feedback loop closes C6

Effort: about one day. Priority: first.

ITEM	MECHANISM TO IMPLEMENT	PHYSICAL JUSTIFICATION
PM10	Apply the same box-model concentration response, damped by an efficiency factor below one.	The box model is a pure dilution argument valid for any conserved aerosol mass. PM10 is coarse-mode dominated with a higher deposition velocity and shorter residence time, so its amplification under a collapsing mixing depth should be real but smaller than PM2.5's. The factor is documented and bounded, not fitted.
NO _x , path 1	Dilution: the same concentration response.	NO ₂ is a surface-emitted primary pollutant, so a shallower mixing depth concentrates it exactly as it does particulate mass.
NO _x , path 2	Reduced photolytic loss: scale NO ₂ by the J(NO ₂) attenuation ratio.	J(NO ₂) governs the rate at which NO ₂ is split. Attenuating the ultraviolet slows that loss, so NO ₂ accumulates. Both paths act in the same direction under haze, which is a check on the implementation.
O ₃	Recompute photolysis from the <i>converged</i> aerosol optical depth and re-evaluate the ozone response inside the iteration.	Currently photolysis is computed once from baseline aerosol before the solver runs. Moving it inside closes the chemistry side of the loop and makes ozone a genuine participant rather than a downstream consumer.

Table 12.1 After this, all four named species participate and clause C6 is met in full. Each new response needs its own clipped range and its own literature gate entry, on the same pattern as the existing five.

Acceptance: the literature gate grows from five checks to eight or nine, all passing; the ablation in §12.3 is re-run across all four species; and no head regresses.

12.2 Make the hosted instance genuinely fresh closes C7

Effort: about two hours. Priority: second, and it costs nothing.

There is a workaround, and it is better than paying for a larger instance. The constraint is that the *web service* has 512 MB. Nothing requires the pipeline to run on the web service.

1. A **scheduled GitHub Actions workflow** runs the full pipeline on a hosted runner. Standard runners provide four cores and 16 GB of RAM, against a measured peak of 1.1 GB, so the pipeline runs at full resolution with an order of magnitude of headroom.
2. The workflow writes the snapshot bundle and **commits it back to the repository**.
3. Render **auto-deploys on push**, so the hosted site begins serving the fresh bundle within a couple of minutes.
4. Public repositories get **unlimited free Actions minutes**, so a three or six hourly refresh costs nothing.

Why this is not a compromise

No operational forecast system re-solves its physics when a user loads a page. NWP runs on a cycle and serves the most recent cycle's output. A scheduled refresh is therefore not an approximation of real time operation; it *is* how real time operation works. The remaining artefact is that the free web service still sleeps after fifteen minutes idle, so the first request after a quiet period is slow. That is a cold start, not a stale forecast, and the two should not be confused.

A scheduled workflow on a repository with no commits is disabled after sixty days of inactivity. This workflow commits on every run, so it keeps itself alive.

A second, complementary route is to cut the peak memory itself: process the forecast in cell batches writing incrementally, hold the panel in float32 throughout, and release the training panel before the coupling stage. That would plausibly bring the pipeline under 512 MB, but it is real engineering effort and the scheduled route delivers freshness *and* keeps full resolution today.

12.3 Score the coupling ablation closes §11.4

Effort: a few hours, mostly compute.

Re-run `scripts/validate.py` without `--skip-ablation` on the current multi winter models, and report the result split by season, aerosol band and ventilation class so that "where it helps" is answered with a table rather than a single flat number. A regime-resolved negative result is far more informative than an aggregate one, because the feedback is physically expected to do nothing on clean afternoons and the aggregate dilutes the winter signal with them.

12.4 A like for like comparison against the DSS

Effort: one to two days, dependent on archive access.

Re-run the hindcast on archived *forecast* meteorology rather than ERA5 reanalysis, so that both systems are asked to predict the weather as well as the chemistry. Until then Table 8.3 is presented with its caveat and is not claimed as a skill result.

12.5 Beyond the statement

- **Multi layer chemistry.** Resolving two or three layers rather than one would let the plume arrive aloft and mix down explicitly rather than through a gate.
- **TROPOMI derived VOC regime map.** Formaldehyde to NO₂ ratio, with the threshold near 3.1, would let the ozone chemistry be regime aware.
- **Assimilation of station observations** into the initial state rather than only into the feature set.

ITEM	EFFORT	CLOSES	COST	RESULT
12.1 Species into the loop	~1 day	C6	none	All four named species coupled
12.2 Scheduled refresh	~2 hours	C7	none	Hosted site genuinely current
12.3 Ablation	~3 hours	\$11.4	none	Evidence the coupling earns its place
12.4 Like for like DSS	1 to 2 days	\$8.3 caveat	archive access	A defensible skill comparison

Table 12.2 After 12.1 and 12.2, ten of the eleven clauses are met in full and C1 remains partial by construction, because we consume a coupled model rather than run one.

A Appendix: where every number lives

No figure in this report was typed by hand. Each was read from the file named below, and each of those files is produced by a script in the repository.

CLAIM	FILE	PRODUCED BY
Twelve head skill, Table 7.2	models/metrics_multiwinter.json	scripts/train.py
Leave one station out, Table 8.1	models/loso.json	scripts/loso.py
Coupling magnitudes, Table 4.1	data/snapshot/coupling.json	vayuchakra/feedback.py
Photolysis and ozone, Table 4.2	models/ozone_sensitivity.json	scripts/ozone_sensitivity.py
Plume calibration, Table 6.1	models/plume_calibration_octnov.json	scripts/plume_calibrate.py
GRAP probabilities, Table 8.2	models/uncertainty_pm25_24h.json	scripts/train_uncertainty.py
DSS head to head, Table 8.3	models/validation.json	scripts/validate.py
Grid construction, Table 3.1	data/snapshot/health.json	vayuchakra/grid.py
Species in the loop, Table 11.1	vayuchakra/feedback.py	source inspection
Abandoned UV check, §8.4	vayuchakra/photolysis.py	cloud_attenuation_check()

Table A.1 Every claim traced to its artefact.

A.1 Running it

- `pytest` runs 76 tests. They include a mesh walk asserting the two grid tiers leave no coverage gap, and they gate on every literature bound in Table 4.1.
- `DECISIONS.md` records 59 decisions with the measurement behind each, negative results included.
- `scripts/run_all.sh` runs the pipeline end to end.
- The dashboard reads the same API this report cites, so anything on screen traces back the same way.

A.2 Sources cited

- Xing et al., *Impacts of aerosol-photolysis interaction and aerosol-radiation feedback on surface-layer ozone in North China*, Atmos. Chem. Phys. 22, 4101 (2022).
- Nelson et al., *Avoiding high ozone pollution in Delhi, India*, Faraday Discussions 226 (2021), APHH-India.
- *The impact of aerosols on photolysis frequencies and ozone production in Beijing 2012 to 2015*, Atmos. Chem. Phys. 19, 9413 (2019).
- Master Chemical Mechanism v3.3.1 photolysis parameters, University of York.
- *Assessing CAMS reanalysis AOD over India*, Environ. Sci. Pollut. Res. (2025).
- MoES and IITM WRF-Chem Decision Support System (JAMES). Third party output, cited and never redistributed.
- CPCB National Air Quality Index, published breakpoint table.